
3.2 Case Study: Landsat Image-Housing Data Integration for LULC Classification in Indianapolis

The objectives of this case study is to improve urban LULC classification accuracy by incorporating ancillary data (specifically census data) and Landsat imagery at different classification stages and to develop a suitable procedure for effective integration of these two types of data.

3.2.1 Study Area

The study area is located at 39°46'N and 86°09'W, Marion County (the city proper of Indianapolis), Indiana (Fig. 3.1). According to the U.S. Census Bureau, the county has a total area of 1044 km², including 1026 km² of land and 18 km² of water. The average annual temperature is 11.7°C, with the highest temperature of 24.6°C in July and the lowest temperature of -1.9°C in January. Annual precipitation is evenly distributed throughout the year, and average annual rainfall is 1021 mm, with the least amount of precipitation occurring in February. It is the core of the Indianapolis metropolitan area. The county seat—Indianapolis—was called “plain city” because of its flat topography (elevation ranges from 218 to 276 m above sea level), which provides the possibility of expansion in all directions. In recent decades, the city has been experiencing areal expansion through encroachment on agricultural land and other nonurban land owing

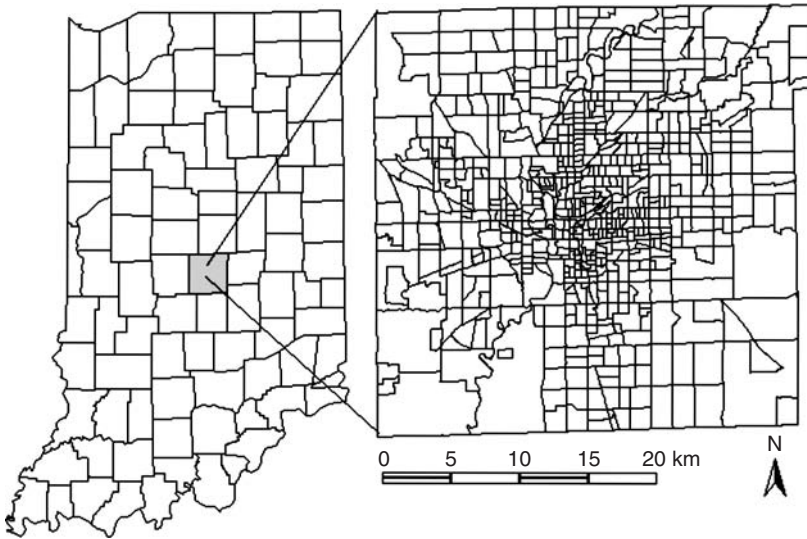


FIGURE 3.1 Study area—Marion County, Indiana.

to population increases and economic growth (the population of Marion County increased by approximately 7.9 percent from 1990 to 2000).

According to the 2000 Census, there were 860,454 people with a density of 838/km². There were 387,183 housing units at an average density of 377/km². With its large population, Indianapolis ranks as the twelfth largest city in the United States. In 1999, the median household income in the county was \$40,421, and the per capita income was \$21,789, with 11.40 percent of the population and 8.7 percent of families below the poverty line.

Indianapolis has the highest concentration of major employers and manufacturing, professional, technical, and educational services in the state. With its moderate climate, rich history, excellent education, and abundant social services, arts, leisure, and recreation, Indianapolis was named as one of America's "Best Places to Live and Work" (*Employment Reviews*, August 1996). In 1996, it ranked fifth on the list of the 30 best cities for small business by *Entrepreneur* magazine, and it was one of the top 10 metropolitan areas in the nation for business success in a 1996 study by Cognetics, a research firm in Cambridge, Massachusetts.

3.2.2 Datasets Used

The primary data sources used in this research are Census 2000 data and Landsat Enhanced Thematic Mapping Plus (ETM+) images (Fig. 3.2). The Landsat 7 ETM+ image (L1G product of path 21, row 32)

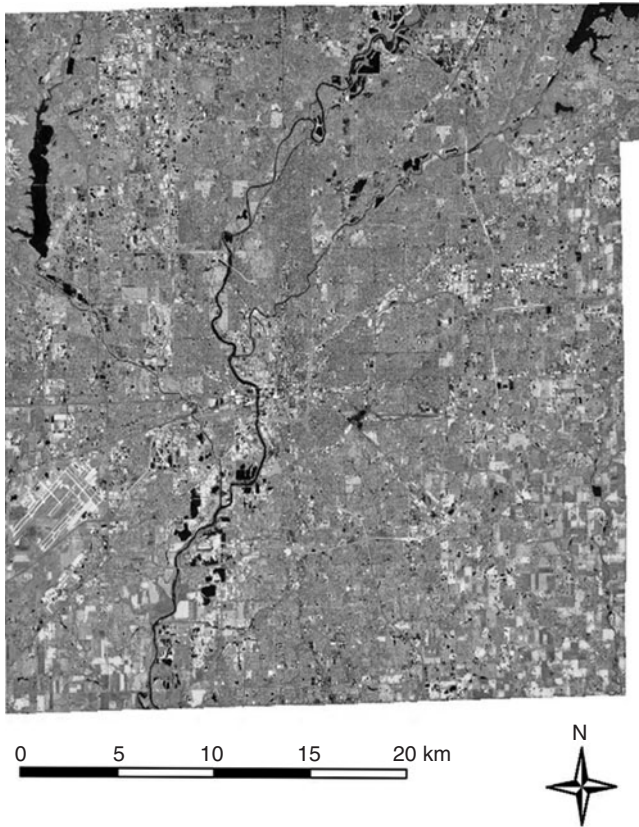


FIGURE 3.2 Landsat ETM+ image (bands 4, 5, and 3 as red, green, and blue, respectively) of Indianapolis acquired on June 22, 2000. See also color insert.

used in this study was acquired on June 22, 2000, under clear-sky conditions. The data were converted radiometrically to at-sensor reflectance using an image-based correction method (Markham and Barker, 1987). Although the L1G ETM+ data were corrected geometrically, the geometric accuracy was not high enough for combining them with other high-resolution datasets. Hence the image was further rectified to a common Universal Transverse Mercator (UTM) coordinate system based on 1:24,000 scale topographic maps. A root-mean-square error of less than 0.5 pixels was obtained in the rectification.

Census 2000 data from the U.S. Bureau of the Census include (1) tabular data stored in Summary File 1, which contains information about the population, families, households and housing unit, etc. and (2) spatial data called topologically integrated geographic encoding and referencing (TIGER) data that contains data representing the position and boundaries of legal and statistical entities. These two

types of data are linked by Census geographic entity code. The U.S. Census has a hierarchical structure consisting of 10 basic levels: United States, region, division, state, county, county subdivision, place, Census tract, block group, and block. For population estimation, the block-group level was selected as the work level, whereas the block level was used as the work level for classification. Other ancillary data used in this research included (1) aerial photographs taken in the summer 2003 that were downloaded from the Indiana Spatial Data Portal and (2) GIS zoning data that were created by an Indianapolis and Marion County GIS team with the state plane coordinate system. All data, including housing, population, aerial photograph, zoning, and ETM+ images, were georegistered to UTM coordinates before data integration.

3.2.3 Methodology

Landsat ETM+ image, zoning, and Census housing data were combined for use in LULC classification in this research. The strategy of the LULC classification procedure is illustrated in Fig. 3.3. Zoning and housing data were used in different stages of image classification (e.g., pre-classification, during classification, and post classification) to identify a suitable procedure for improving LULC classification accuracy.

Housing information was extracted from the Census data at block level, and housing density (Fig. 3.4) was calculated by dividing housing units by block area. In order to incorporate housing information into the Landsat ETM+ image, housing data must be converted to raster format. Two methods are often used to this conversion. One method is to conduct rasterization, and the other is to generate a

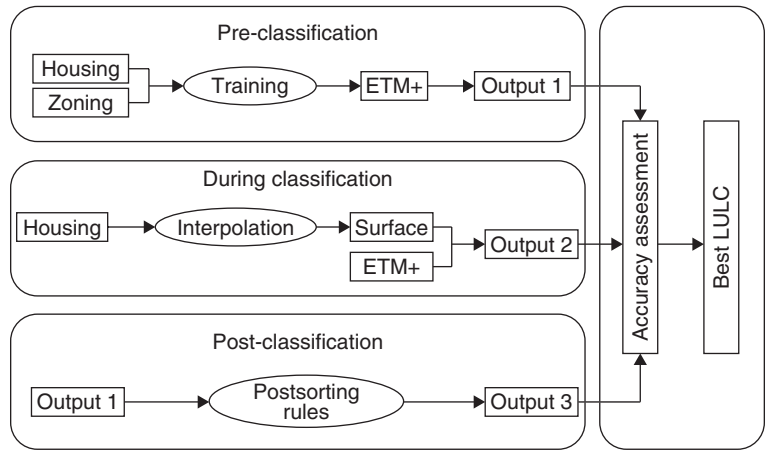


FIGURE 3.3 Strategies of LULC classification by incorporating zoning and housing datasets for improving classification performance.

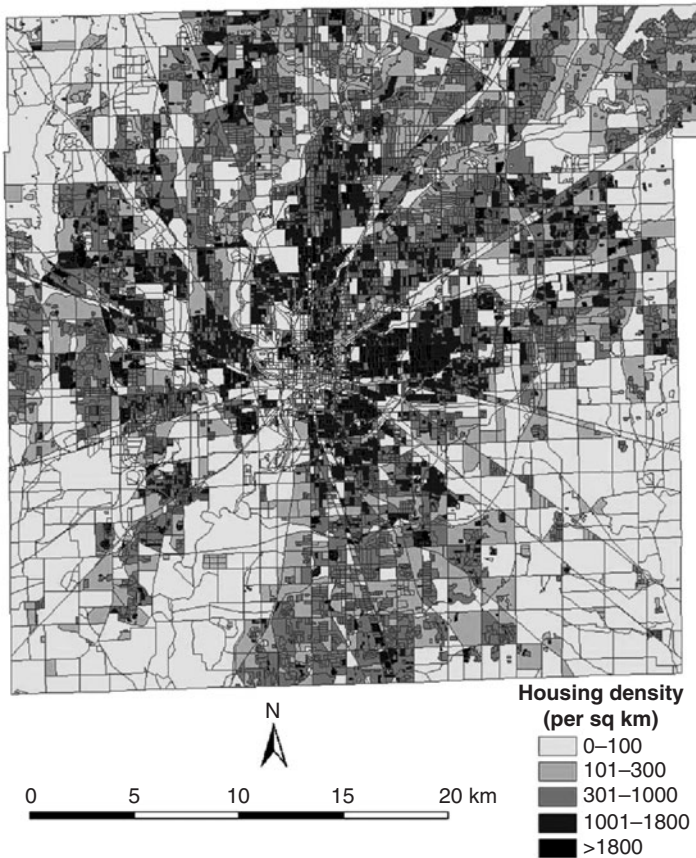


FIGURE 3.4 Housing density distribution at Census block, Marion County, Indiana, in 2000. See also color insert.

housing surface by using spatial interpolation algorithms. In this study, the centroids of blocks (Fig. 3.5) were identified with the aid of ArcGIS, and then inverse-distance-weighting (IDW) interpolation was used to generate a housing surface.

The smooth and continuous housing surface derived from the centroids of blocks using IDW (Fig. 3.6) enabled conversion of spatial data from irregular zonal units (blocks) into regular units (surface cells or pixels) that had a similar format as the ETM+ image. One important characteristic of this model is that it preserves the total housing units (Mesev, 1998). In contrast, the rasterized housing density (Fig. 3.7) was not continuous; cell values were the same within a block, but it preserved most of the original housing density of the blocks. Both were integrated with the ETM+ image as extra layers by stacking them on Landsat ETM+ six bands for further classification.

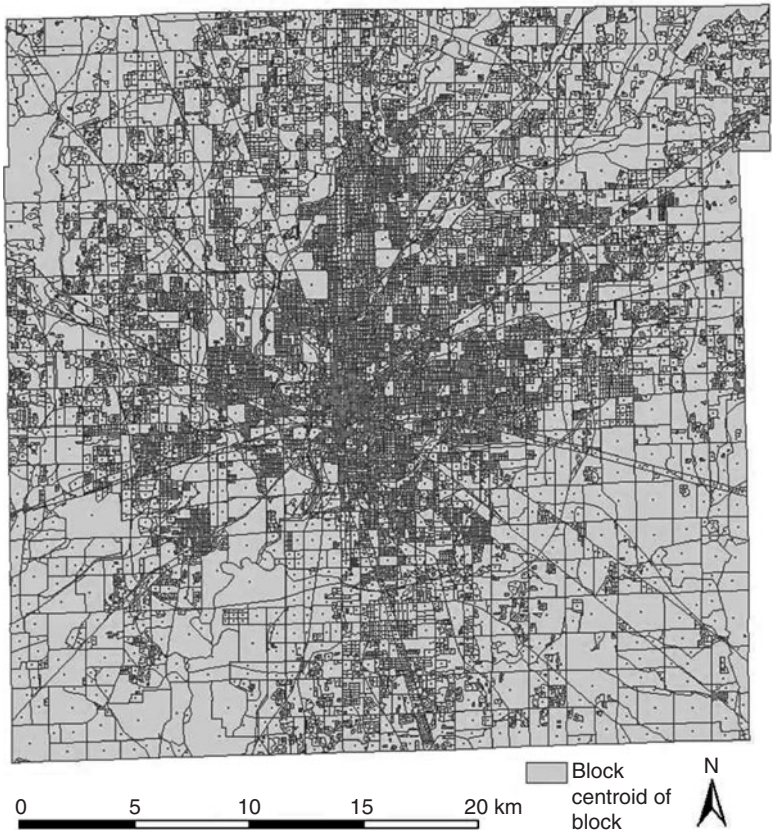


FIGURE 3.5 Centroids of Census block. See also color insert.

In the Pre-Classification Stage

A traditional supervised classification (specifically, the maximum-likelihood classifier) was used to classify the ETM+ image using its light-reflective bands. Before implementing image classification, selection of good-quality training samples is critical. In this research, housing density, zoning data, and high spatial-resolution aerial photographs were used to assist the identification of training samples of different urban LULC classes. The ETM+ image was classified initially into 11 classes (i.e., commercial, transportation, industrial, water, low-density residential, medium-density residential, high-density residential, grass, crop land, fallow, and forest). Figure 3.8 provides examples of typical LULC data appearing on the aerial photograph. The 11 classes then were merged to 8 classes by combining commercial, transportation, and industrial as urban and combining crop and fallow as agricultural land.

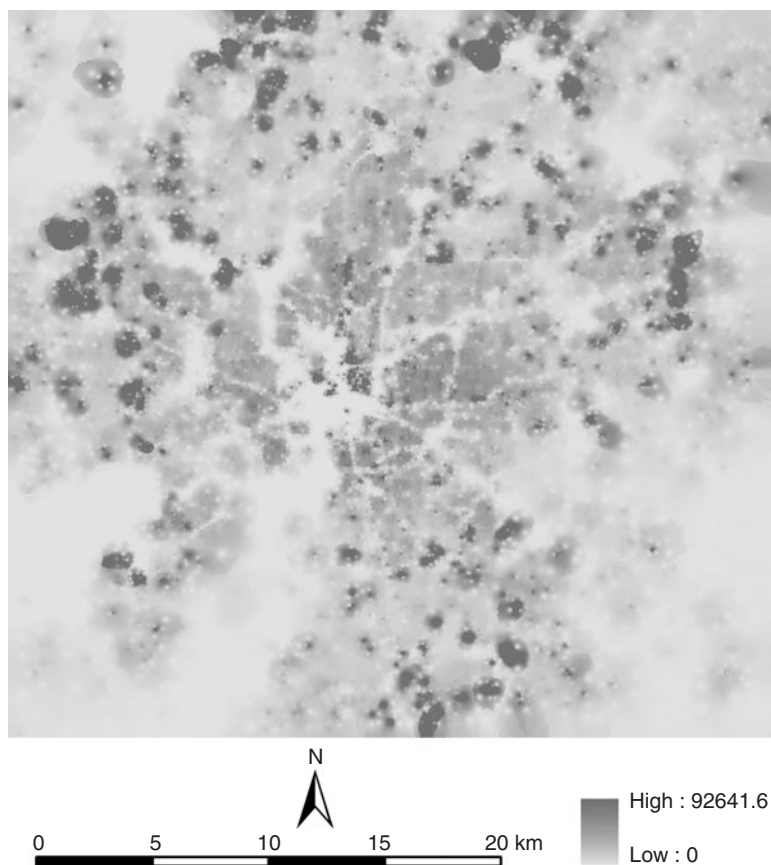


FIGURE 3.6 Housing surface generated by IDW interpolation. See also color insert.

For residential density, there are no agreed-on standards for what constitutes high, medium, and low density. A high density in Indianapolis might be medium or even low density in Shanghai. Therefore, in this study, the criteria for separating different densities of residential lands were adopted based on housing density and zoning data. For example, residential lands having housing units of more than 1300 per square kilometer were assigned as high-density residential areas, those having housing units of fewer than 400 per square kilometer were assigned as low-density residential areas, and those having housing units between 400 and 1300 per square kilometer were assigned as medium-density residential areas. Table 3.1 gives the definitions for these eight classes. This classification result was used as a base map for the post-classification sorting.

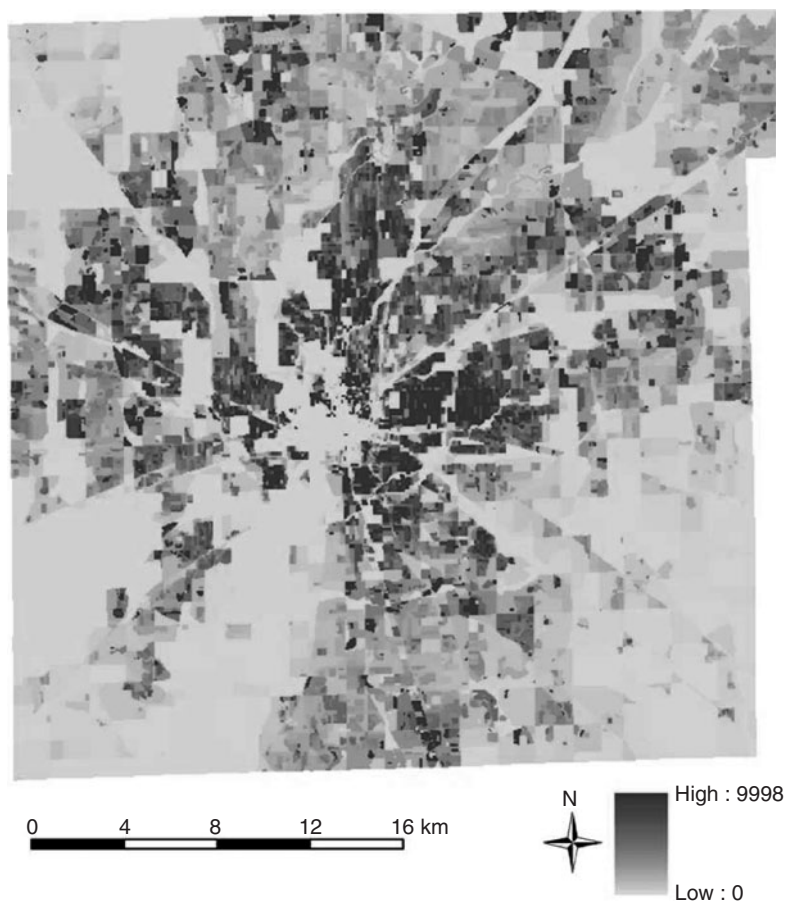


FIGURE 3.7 Rasterized housing density. See also color insert.

During the Classification

In this stage, the housing dataset was incorporated as an extra layer into the ETM+ image. Two housing data layers—rasterized housing density and housing surface—were combined with the ETM+ image as additional channels for image classification. Supervised classification with the maximum likelihood algorithm then was used to classify the combined image.

In the Post-Classification Stage

Because of similar spectral characteristics in certain LULC types, such as between high-density residential and commercial areas and between low-density residential and forest areas, urban LULC classification is often difficult based on spectral signatures (Lu and Weng, 2004). Thus, in this research, housing density was used to correct the pixels that were misclassified during the classification procedure. Based on the

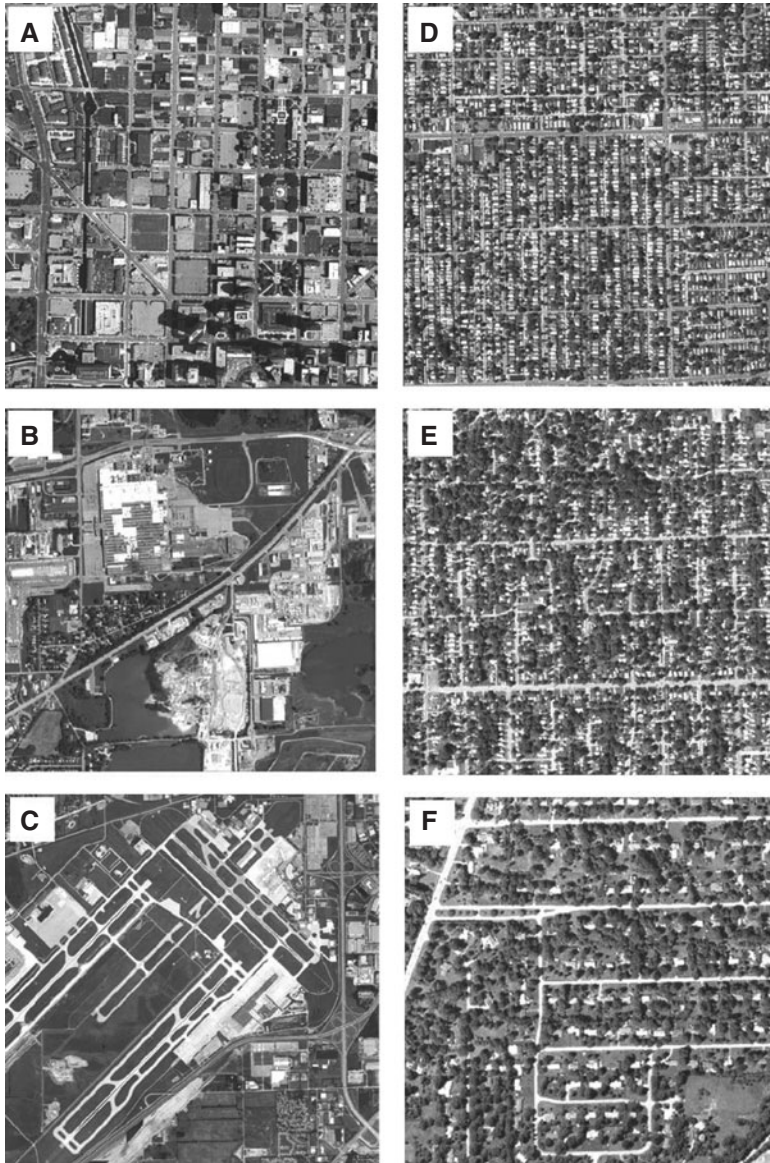


FIGURE 3.8 Examples of typical land use: (a) commercial, (b) industrial, (c) transportation, (d) high-density residential, (e) medium-density residential, (f) low-density residential. See also color insert.

housing density distribution at the block level, the following rules indicated in Table 3.2 were developed. Finally, for convenience, grass, crop land, and forest areas were merged into a new group called *vegetation*. By applying the rules shown in the table, the problematic pixels could be merged into the categories most appropriate for this research.

LULC Type		Definition
Urban		Commercial, transportation, and industrial
Residential	High-density residential	Residential areas having a housing density of greater than 1300 per square kilometer
	Medium-density residential	Residential areas having a housing density of greater than 400 and fewer than 1300 per square kilometer
	Low-density residential	Residential areas having a housing density of fewer than 400 per square kilometer
Crop land		Herbaceous vegetation that has been planted or is intensively managed for the production of food, including croplands such as corn, wheat, and soybean, as well as fallow land
Water		All areas of open water, including lakes, rivers, and streams and ponds
Forest		The areas covered by trees, including natural deciduous forest, evergreen forest, mixed forest, and shrubs
Grass		The areas covered by herbaceous vegetation, including pasture/hay planted for livestock grazing or the production of hay; also includes the urban/recreational grasses, such as parks, lawns, golf courses, airport grasses, etc

TABLE 3.1 Definitions of Land Use and Land Cover Types

Condition		Modification
From LULC Type	Housing Density	To LULC Type
Urban	>1300	High-density residential
High-density residential	<50	Urban
Medium-density residential	<50	Urban
High-density residential	>400 and <1300	Medium-density residential
Medium-density residential	<400	Low-density residential
Low-density residential	>400 and <1300	Medium-density residential
Low-density residential	<50	Vegetation

TABLE 3.2 Decision Rules Developed for Postsorting

3.2.4 Accuracy Assessment

In LULC classification, accuracy assessment of the classification results is often required (Foody, 2002). Many approaches, such as overall accuracy, producer's accuracy, user's accuracy, and kappa coefficient, have been used for evaluating classification accuracy. Their meanings and calculations have been described extensively in the literature (Congalton, 1991; Congalton and Mead, 1983; Hudson and Ramm, 1987; Smits et al., 1999). In reality, the most frequently used method for quantitatively analyzing LULC classification accuracy may be the error matrix, and thus it was used in this study. The accuracy assessment for the three classification images was conducted with a randomly sampling method. Fifty samples for each LULC type were selected. The reference data were collected from high-spatial-resolution aerial photographs. Overall accuracy, producer's accuracy, user's accuracy, and kappa statistic were calculated based on the error matrices.

3.3 Classification Result by Using Housing Data at the Pre-Classification Stage

Supervised classification using maximum likelihood classification was performed with all six ETM+ reflective bands. Training samples were selected with the assistance of housing-density and zoning GIS data, as well as aerial photographs. Owing to high variability within the same LULC types, training samples were selected to be as detailed as possible. For example, transportation, central business district (CBD), and industrial classes vary considerably in spectral response, although they were classified as one class in the final result. Therefore, training samples were selected separately based on different types of urban land use. A similar situation happened with agricultural lands (lands planted with crops and lands lying fallow). Because of the spectral confusion problem, the image was classified initially into 11 categories. The cell-array table of separabilities (Table 3.3) indicated that separability between vegetation types (crop land and forest), between residential intensity levels, between high-density residential and transportation areas, and between low-density residential and grass/crop areas was very low owing to the spectral similarity of these features, which may result in poor classification accuracy. Figure 3.9 shows the classification image, in which the 11 categories were merged to 8 LULC classes (i.e., water, urban, high-density residential, medium-density residential, low-density residential, forest, grass, and crop land). On this image, some obvious errors were found. For example, the commercial area in the northwest (*a*) was mixed with the medium-density residential area. Medium-density residential area also invades into the airport (*b*) and industrial areas (*c*).

Signature Name	2	3	4	5	6	7	8	9	10	11
Industrial (1)	2000	2000	2000	2000	2000	2000	2000	2000	2000	2000
Crop land (2)		2000	1982	2000	2000	2000	2000	2000	2000	1970
Fallow (3)			2000	2000	2000	2000	2000	2000	2000	2000
Forest (4)				2000	2000	2000	2000	2000	2000	2000
Grass (5)					2000	2000	2000	2000	2000	1985
Transport (6)						2000	2000	1482	1931	1997
Commercial (7)							2000	2000	2000	2000
Water (8)								2000	2000	2000
Residential-H (9)									1239	1763
Residential-M (10)										820
Residential-L (11)										0

TABLE 3.3 Separability Cell-Array for Six ETM+ Bands Using Transformed Divergence (Best Average Separability: 1957.1 with Combination 1, 2, 3, 4, 5, 6)

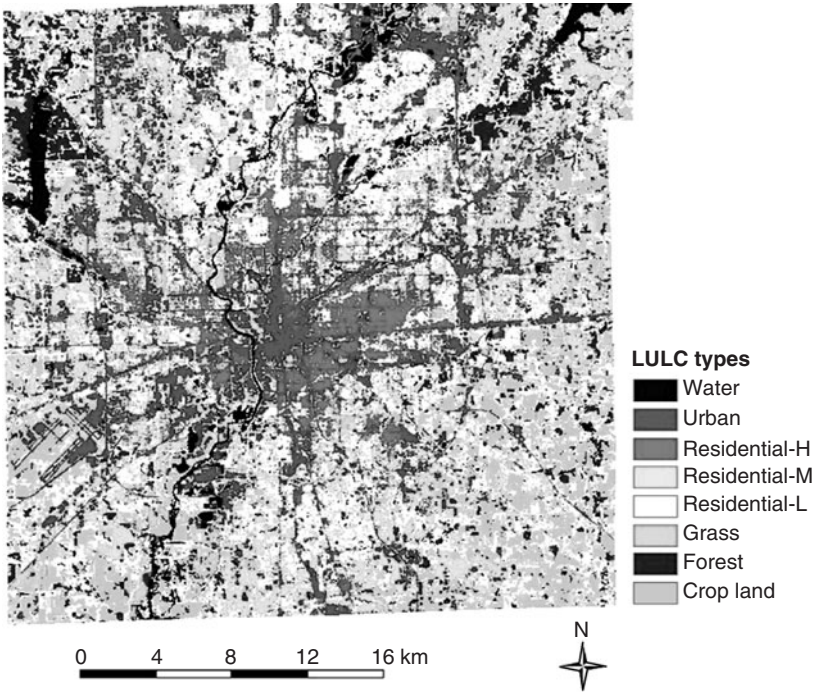


FIGURE 3.9 LULC classification image based on intergration of housing and the ETM+ image at the pre-classification stage. See also color insert.

However, medium-density residential areas supposed to appear in areas adjacent to high-density residential areas were classified as low-density residential areas (*d*). Table 3.4 shows the classification accuracy based on the eight LULC types. An overall accuracy of 70.5 percent and a kappa value of 0.6629 were achieved. The accuracies of the residential types were the lowest. Confusion occurred between urban and high-density residential areas; between high-density residential and medium-density residential areas; between medium-density residential and low-density residential areas; between low-density residential, grass, and forest areas; and between grass and crop land areas. These confusions were the major factors contributing to the low accuracy of image classification. Mixed pixels, such as mixtures of impervious surface, forest, and grass in the low-density residential areas, were another factor for low classification accuracy. Selecting training samples for low-density residential land was difficult even with the assistance of zoning and housing data.

	Pre-classification		During classification		Post-classification	
LULC	PA	UA	PA	UA	PA	UA
Water	100.00	96.00	98.00	98.00	97.92	95.92
Urban	61.19	82.00	75.47	80.00	86.57	79.45
Residential-H	70.97	44.00	73.47	72.00	51.61	72.73
Residential-M	46.51	40.00	56.10	46.00	81.40	74.47
Residential-L	65.79	50.00	60.00	36.00	76.32	64.44
Grass	56.96	90.00	60.53	92.00	65.82	78.79
Crop land	79.49	62.00	82.35	56.00	76.92	61.22
Forest	90.91	100.00	74.63	100.00	89.09	100.00
Overall accuracy	70.50		72.50		79.00	
Overall kappa	0.6629		0.6857		0.7574	

PA = producer's accuracy; UA = user's accuracy.

TABLE 3.4 Comparison of Different Classification Schemes

3.4 Classification Result by Integrating Housing Data during the Classification

Ancillary data also can be used as extra channels by combining them with remotely sensed data. In this study, rasterized housing density and a housing surface developed with the IDW interpolator were integrated into the ETM+ image as additional layers. A supervised classification was applied to classify the combined dataset. It was found that incorporation of rasterized housing density failed to identify some LULC classes, such as forest, crop, and grass. The integration of housing surface layer with the ETM+ image did improve the average separability to a certain degree (separability value from 1957 to 1984). A comparison of Table 3.3 with Table 3.5 indicates that the separability between any pair of LULC types mentioned in the last section was increased. Table 3.4 shows that the accuracies of high-density residential and medium-density residential were improved, but the accuracies of low-density residential, urban, and crop land decreased. Both overall accuracy and the kappa value increased, although not significantly. Figure 3.10 shows the classified image based on the during-classification approach.

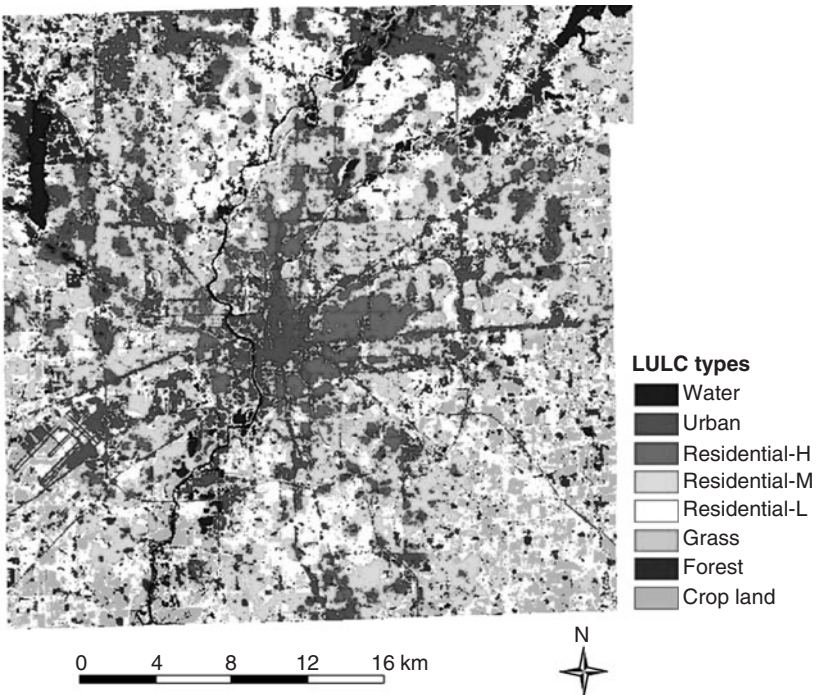


FIGURE 3.10 LULC map based on integration of housing surface into the ETM+ image as an additional layer (during the classification). See also color insert.

Signature Name	2	3	4	5	6	7	8	9	10	11
Industrial (1)	2000	2000	2000	2000	2000	2000	2000	2000	2000	2000
Crop land (2)		2000	1982	2000	2000	2000	2000	2000	2000	1992
Fallow (3)			2000	2000	2000	2000	2000	2000	2000	2000
Forest (4)				2000	2000	2000	2000	2000	2000	2000
Grass (5)					2000	2000	2000	2000	2000	1998
Transp. (6)						2000	2000	1770	1944	1999
Commercial (7)							2000	2000	2000	2000
Water (8)								2000	2000	2000
Residential-H (9)									1641	2000
Residential-M (10)										1578
Residential-L\ (11)										0

TABLE 3.5 Separability Cell-Array for ETM/Housing Surface Using Transformed Divergence (Best Average Separability: 1983.69 with Combination 1, 2, 3, 4, 5, 6, 7)

3.5 Classification Result by Using Housing Data at the Post-Classification Stage

The ancillary data can be used further for postprocessing of the classified images. In this research, the LULC image (see Fig. 3.9) developed from the ETM+ image was used as a base. Postsorting rules were applied to this image to produce a refined LULC image (Fig. 3.11). Because the objective of using these rules was to refine urban and residential land uses, they did not have much impact on water, agricultural land, and forest in terms of accuracy. However, the post-classification sorting had a great impact on the residential and urban areas, as well as grasslands. Table 3.4 indicated that the overall accuracy and kappa value were increased to 79 percent and 0.7574 respectively, compared with 70.5 percent and 0.6629 before the sorting. The accuracies of residential land areas were improved considerably, but the accuracies of urban areas and grasslands were decreased to varying degrees. Also, the obvious errors mentioned in the pre-classification stage disappeared. A comparison of the three classification schemes indicates that integration of housing data through postsorting is more effective than through adding housing surface as a layer into the dataset to be classified.

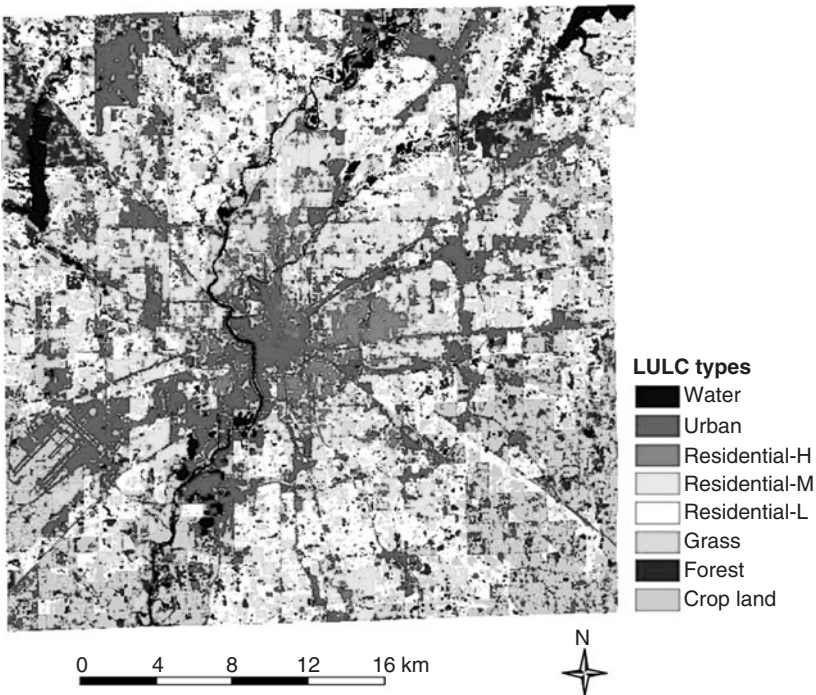


FIGURE 3.11 LULC map based on integration of housing data at the post-classification stage. See also color insert.

3.6 Summary

This chapter examined the role of Census housing data in LULC classification for an urban area through integration with the Landsat ETM+ image at different classification stages based on a case study of Indianapolis. The results indicate that use of housing data in the post-classification stage provides a better result than use of such data during the classification stage as an additional layer.

In urban areas, an important data source is housing data. However, the use of the housing data has not fully been examined in urban LULC classification. This research examined the use of housing data at different stages of image classification, that is, for assisting in the selection of training sample plots at the pre-classification stage, for combination with the remote sensing image as an extra channel at the during-classification stage, and for modifying the classified image through use of expert rules at the post-classification stage. When housing data were used as an extra channel during the classification procedure, two factors may contribute to their ineffectiveness for improving classification accuracy. One possible factor is the abnormal distribution of housing data in the training samples. Because the assumption of a normal distribution of the training samples was not satisfied, the effectiveness of the maximum likelihood classifier, if employed, may suffer. Another factor is associated with the relatively coarse spatial resolution of the interpolated housing surface image, which was developed from Census data at the block level. In this study, the control points used for generating housing surface were geometric centers, that is, centroids of the blocks, which often introduced errors when used to represent the source zones. Calculation of the centroid of an area depends on the coordinates of points defining the boundaries of a source zone. If a source zone is simple or symmetric, the centroid would be a good control point, and the estimated value for a grid cell would be more reliable. However, if a zone is irregular and complicated, the location of centroid can be significantly affected, and the interpolation result may be biased. In reality, Census blocks are often not in regular shapes. The nonuniform distribution of housing units within a block can further complicate the problem. Another important issue is how to select an appropriate spatial interpolation algorithm because different interpolators may influence the final result of LULC classification substantially.

Use of housing data in the post-classification stage provided the best results in this research. The deterministic decision rules used to refine the base LULC are critical for their effectiveness; therefore, they need to be developed carefully. In reality, developing the decision rules is a challenge because there are no standard criteria for distinguishing the different residential densities. An analyst needs to carefully check the existing information, such as Census data and historic land use, and information collected by local planning agencies. In this

research, decision rules were developed based on housing density distribution with zoning data as a reference, but the break points for housing density seemed to be somewhat difficult to determine. This may have contributed to unsatisfactory improvement in separating the low-density residential class from the others.

Image classification in urban areas using medium-spatial-resolution (10 to 100 m) data is a challenge task. Although much research has been conducted with different techniques, a comprehensive comparison on which to generate some common guidelines for urban classification is lacking because of the use of different datasets, different image processing procedures, and different complexities of urban landscapes. For this same study area, we have conducted research with different approaches. It is pertinent to make a comparison and to arrive at some general observations.

Lu and Weng (2005) compared eight different image processing routines to identify suitable remote sensing variables for urban classification for the same study area by using the ETM+ image. Incorporation of spectral signature, texture, and surface temperature was examined, as well as data fusion techniques for combining a higher-spatial-resolution image with lower-spatial-resolution multispectral images. The research indicated that incorporation of texture from lower-spatial-resolution images or of a temperature image cannot improve classification accuracies. However, incorporation of textures derived from a higher-spatial-resolution panchromatic image did improve the classification accuracy. In particular, use of data fusion result and texture image yielded the best classification accuracy. Incorporation of texture and spectral signatures is conventionally assumed to provide better classification results than use of pure spectral signatures (Butusov, 2003; Shaban and Dikshit, 2001). However, the success for incorporating texture images depends on the type of texture measure, the image used to derive textures, and the selection of moving window size (Lu and Weng, 2005).

Owing to the complexity of urban landscapes and the limitations of current sensor technology, per-pixel-based classification approaches have proven to be difficult for urban LULC classification. Recently, subpixel analysis using spectral mixture models, which unmix an image into different fractions, has been demonstrated to improve classification accuracy (Lu and Weng, 2004; Phinn et al., 2002; Rashed et al., 2001). Spectral unmixing provides a more realistic representation of the true nature of a surface compared with that provided by the assignment of a single dominant class to every pixel by statistical models (Campbell, 2002) and is suitable to solve the mixture problem for medium- or low-spatial-resolution data. Lu and Weng (2004) compared a subpixel-based classification approach with traditional supervised classification in this study area and found that the subpixel-based approach can improve classification accuracy significantly (by 9 percent) compared with the maximum likelihood classifier.

Although the use of textures, data fusion of multiresolution images, and subpixel information improved urban LULC classification accuracies, the separation of different densities of residential areas is still difficult in urban LULC classification. Therefore, in another study in the same study area, Lu and Weng (2006) developed a new approach based on the combined use of impervious surface and population density for urban land use classification. Five urban land use classes (i.e., low-, medium-, high-, and very-high-density residential areas and commercial/industrial/transportation uses) were developed with an overall classification accuracy of 83.78 percent. This research indicates that setting up suitable thresholds for impervious surface and population density for each urban land use category is the key to success with this approach.

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